MOBILE DRILL BRAIN

Calibration Technician Training Program

Module 002 - SPT Analyzer Kit, Instrumented Rods, and Job Readiness

v0.2 - Source-Backed + SME Reviewed

Document type	Calibration technician training module
Training lane	Workspace 006-B - Calibration Technician Training
Module focus	Pile Dynamics Inc. SPT Analyzer kit, instrumented rods, cable path, job readiness, file setup, and equipment escalation
Source basis	Calibration Service Module Extraction Worksheet v0.1; Centerpiece/source repository context; SME field notes and equipment review
Status	Internal training draft - ready for review and future trainer validation
Boundary	This module teaches equipment readiness and profile verification. It does not replace official analyzer operation training, final report approval, or senior calibration-tech judgment.

1. Module Purpose

Module 002 teaches a new calibration technician how to prepare, identify, verify, and protect the core equipment used during an SPT automatic hammer calibration job. The module focuses on the Pile Dynamics Inc. SPT Analyzer kit, instrumented rod sections, cable path, rod/anvil readiness, job setup information, and escalation when equipment is damaged or unclear.

Training intent: The trainee should understand what must be present before data collection begins and why missing, damaged, dirty, mismatched, or incorrectly identified equipment can compromise the calibration job.

2. Learning Outcomes

- Identify the required SPT Analyzer kit components for a calibration job.
- Explain the basic cable path from instrumented rod sensors to the analyzer hardware unit.
- Differentiate accelerometers from strain gage bridges at a training level.
- Explain why AWJ/NWJ rod type must be confirmed before the job.
- Describe when and how the analyzer software creates the data file.
- Recognize required jobsite PPE and field workstation needs.
- Know when equipment damage must stop normal workflow and trigger an office call.

3. Module Boundary

This module teaches	This module does not teach
Kit readiness, required equipment, and field packing logic.	Button-by-button analyzer operation.
Rod type, sensor, cable, connector, and power readiness.	Formal report approval or final customer report release.
File setup fields that must be entered before collection.	Advanced hammer repair or engineering diagnosis.
Damage escalation and office/senior-tech communication.	Customer-facing guarantees about pass/fail results, report timing, or job duration.

4. Required Calibration Job Kit

Every calibration job must have the complete SPT Analyzer kit and job-readiness items on site before data collection begins. Missing kit items should be identified before leaving for the job whenever possible.

4.1 Analyzer / Control Hardware

Item	Training note
Pile Dynamics Inc. SPT Analyzer	The hardware unit that has the SPT Analyzer software installed and is used for calibration data collection.
SPT Analyzer software	Used to create the job/rig data file, collect readings, and store data automatically on the hardware.
Manual start/stop clicker button	One click starts data collection. The second click stops data collection.
Foldable keyboard and mouse	Used when the setup allows easier entry and navigation around the analyzer hardware.

4.2 Sensor and Cable System

Item	Purpose / training note
Sensor 4-to-1 cable, commonly called the pigtail	Connects the four sensors on the instrumented rod and merges them into one connector path.
Hardware-to-pigtail cable	Connects the pigtail/sensor path back to the SPT Analyzer hardware unit.
Military-grade 19-pin twist-lock sealed connectors	Used on the cable path. The twist-lock seals protect the electrical path in field conditions and must be seated properly.
Dielectric grease at connector areas	Helps protect connector areas. A rag is used to wipe visible dirt caught by the grease.
Rag	Used to remove visible dirt at connector/contact areas.

Item	Purpose / training note
	Cleaning solution is not part of the standard kit list for this module.

4.3 Instrumented Rod System

Item / concept	Training note
AWJ or NWJ instrumented rod section	Only the scheduled/confirmed rod type is required for the job. Sometimes both rods are needed, but that is decided rig-by-rig during scheduling.
Strain gage bridges	Measure the dynamic force applied to the rod by the hammer blow.
Piezoresistive accelerometers	Measure acceleration of the rod upon impact. The analyzer uses this data to calculate rod velocity.
Backup accelerometers	Carried in case of field damage or faulty accelerometer issue.
Accelerometer tightening hardware	Washer with bolt and wrench. These stay in the kit and are specifically for accelerometer tightening.
Sensor installed-storage practice	Sensors stay installed when not in use because frequent removal and installation can wear the precisely drilled mounting holes in the rods.
Strain gauge damage rule	Strain gauges are built into the instrumented rod body and are not removable or field-replaceable. Suspected damage requires immediate escalation.

4.4 Power, Storage, Tools, Workstation, PPE, and Documentation

Category	Required items / notes
Power and charging	Fully charged battery pack, backup battery pack, battery charger, and vehicle-friendly charging adapter for jobsite use.
Storage and protection	Weatherproof, heavily padded SPT storage case for analyzer kit, cables, connectors, sensors, and accessories.
Tools and backup hardware	Washer with bolt and wrench for accelerometer tightening. Keep the tightening hardware in the kit.
Field workstation	Six-foot folding table. Provides a stable place for analyzer hardware, paperwork, and kit items.
PPE baseline	Hardhat and steel-toe boots. Site-specific requirements always override the Mobile Drill baseline.
Hard-copy documentation	Binder with data collection forms for blow count, LE, LP,

Category	Required items / notes
	depth, and process notes. Black or blue ink pen.

5. Rod Type and Job-Specific Rod Selection

Rod type is not guessed in the field. The scheduled/confirmed rod type controls which instrumented rod needs to go to the job. The calibration schedule and pre-job preparation should determine whether AWJ, NWJ, or both instrumented rods are needed.

Rule	Meaning
Do not bring the wrong rod type.	The calibration setup depends on the rod family being used.
Do not assume both rods are needed.	Bring the rod type required for the rig/job. Bring both only when scheduling confirms both are needed.
Do not assume one rod fits every rig.	Rod need is confirmed on a rig-by-rig basis.

6. Software Setup and Automatic File Creation

File creation happens before data collection. During setup, the technician enters required identification fields into the analyzer software. The software then creates the save file automatically in the background and stores data on the analyzer hardware in the designated automatic data folder.

Required setup field	Why it matters
Customer name	Identifies who the data belongs to.
Rig model	Identifies the rig being calibrated.
Rig serial number	Required for report identity and rig-specific traceability.
Borehole ID	Ties the data to the correct test location/boring.
Driller name	Documents who operated the rig during data collection.
Technician name	Documents who performed the calibration data collection.

When moving to a different rig, the technician enters that rig's information and the software creates a new file for that rig. Do not reuse one rig file for another rig.

Current folder-name note: The exact folder name/path is source-pending. For training purposes, refer to it as the designated automatic data folder. The analyzer hardware desktop shows the folder as one of the desktop icons and the software routes data to it automatically.

7. Pre-Job Kit Verification Workflow

1. Review the scheduled job details: customer, location, rig count, expected rod type, and whether one or more rigs are being calibrated.
2. Confirm whether AWJ, NWJ, or both instrumented rods are required for the job.
3. Open the SPT storage case and verify the analyzer hardware, cables, pigtail, connectors, batteries, clicker, keyboard, mouse, and required accessories are present.
4. Verify batteries are charged and that a backup battery and charger are packed.
5. Inspect visible cable and connector condition. Wipe visible dirt from connector areas with a rag if needed.
6. Verify accelerometer tightening hardware is in the kit: washer with bolt and wrench.
7. Verify hard-copy documentation is packed: binder/forms and black or blue ink pen.
8. Verify hardhat, steel-toe boots, and six-foot folding table are ready.
9. If anything is missing, unclear, or damaged, call the office before proceeding as normal.

8. Common Equipment Mistakes

Mistake	Why it matters	Correct behavior
Wrong rod type packed	The calibration setup may not match the rig/hammer/rod configuration.	Confirm AWJ/NWJ requirement during scheduling and before departure.
Loose or poorly seated connector	Sensor communication can fail or produce bad setup checks.	Inspect and properly seat twist-lock connectors.
Visible dirt left in connector area	Dirt captured by dielectric grease can interfere with clean connection handling.	Wipe visible dirt with a rag.
Battery not charged	Data collection can be delayed or interrupted.	Charge battery and bring backup power.
One rig file used for multiple rigs	Data traceability and report identity can become wrong.	Create a new file by entering each rig's information when switching rigs.
Sensor damage handled independently on site	A damaged system can create unreliable data or require rescheduling.	Call the office immediately and explain severity.
Treating strain gauges like replaceable field sensors	Strain gauges are built into the rod body and cannot be field-replaced.	Escalate immediately if strain gauge damage is suspected.

9. Stop / Escalate Conditions

If any analyzer kit, cable, connector, accelerometer, or instrumented rod component is damaged, the technician must call the office immediately and explain the severity.

A standard technician does not independently decide on site whether to continue, abort, or reschedule. That decision must be made with office or senior calibration tech input.

A senior calibration tech may make immediate field judgment when necessary, but the issue still must be communicated to the office.

Condition	Required action
Analyzer hardware damaged or not powering correctly	Call office; do not continue normal setup until directed.

Condition	Required action
Cable/pigtail/connector damage	Call office; explain which component and severity.
Accelerometer damaged, faulty, or loose beyond normal tightening	Use backup only when appropriate and communicate issue; call office when uncertain.
Strain gauge damage suspected	Call office immediately; treat rod as potentially unusable for calibration.
Wrong rod type discovered on site	Call office; do not improvise unsupported setup.
Data file identity entered wrong	Correct before collection if caught; if discovered after collection, document and escalate before report handoff.

10. Field Profile Checklist Before Data Collection

Check item	Complete?
Customer name entered correctly	[ ]
Rig model entered correctly	[ ]
Rig serial number entered correctly	[ ]
Borehole ID entered correctly	[ ]
Driller name entered correctly	[ ]
Technician name entered correctly	[ ]
Correct rod type confirmed	[ ]
Pigtail/cable path connected and seated	[ ]
Battery power available with backup	[ ]
Hard-copy forms and pen ready	[ ]
PPE baseline met: hardhat and steel-toe boots	[ ]
Data is being stored to the designated automatic data folder	[ ]

11. Scenario Practice

Scenario: A technician arrives for a one-rig calibration. Scheduling said the rig uses AWJ, but the customer says the rig may have been changed to NWJ. The AWJ instrumented rod is in the kit. The NWJ rod is not. The customer wants to proceed because the crew is ready.

Expected trainee response:

- Do not assume the AWJ rod is acceptable if the rod type is uncertain.

- Stop normal setup and call the office or senior calibration tech for direction.
- Explain that the correct rod type is required for calibration setup and report traceability.
- Document the uncertainty and any customer-provided information.
- Do not independently decide to continue with an unsupported setup.

12. Knowledge Check

#	Question	Correct answer
1	What is the name of the analyzer hardware used for SPT calibration?	Pile Dynamics Inc. SPT Analyzer.
2	What does the sensor 4-to-1 cable, commonly called the pigtail, do?	It connects the four sensors on the instrumented rod and merges them into one connector path.
3	What connector type is used on the sensor/pigtail/hardware cable path?	Military-grade 19-pin twist-lock sealed connectors.
4	What does the manual start/stop clicker do?	One click starts data collection and the second click stops data collection.
5	What do strain gage bridges measure?	The dynamic force applied to the rod by the hammer blow.
6	What do piezoresistive accelerometers measure?	Rod acceleration upon impact, which the analyzer uses to calculate rod velocity.
7	Can strain gauges be replaced in the field?	No. They are built into the instrumented rod body and are not field-removable or field-replaceable.
8	What backup sensor should be carried for a faulty or damaged sensor event?	Backup accelerometers.
9	What hardware is kept in the kit specifically for accelerometer tightening?	Washer with bolt and wrench.
10	Should both AWJ and NWJ instrumented rods go on every job?	No. Bring the scheduled/confirmed rod type; sometimes both are needed, but that is decided rig-by-rig during scheduling.
11	When is the calibration data file created?	Before data collection, during analyzer software setup.
12	Name the required setup fields used to create the correct save file.	Customer name, rig model, rig serial number, borehole ID, driller name, and technician name.
13	Where is the data stored?	Automatically on the analyzer hardware in the designated automatic data folder.
14	What is the standard PPE baseline for Mobile Drill calibration work?	Hardhat and steel-toe boots unless the jobsite requires more.
15	What should the technician use to remove	A rag. Visible dirt caught by dielectric

#	Question	Correct answer
	visible dirt from connector areas?	grease should be wiped away.
16	What should a standard technician do if analyzer kit, cable, connector, accelerometer, or instrumented rod damage is found?	Call the office immediately and explain severity. The standard tech does not independently decide to continue, abort, or reschedule.

Pass / Review Standard

- Passing score: 85% or higher.
- Automatic trainer review if trainee misses strain gauges are not field-replaceable.
- Automatic trainer review if trainee misses that file setup happens before collection.
- Automatic trainer review if trainee misses that rod type must be confirmed before the job.
- Automatic trainer review if trainee misses that damaged equipment must be reported to the office.
- Automatic trainer review if trainee misses the clicker start/stop function.

13. Trainer Signoff Checklist

Trainer check	Trainer initials / notes
Trainee can identify the Pile Dynamics Inc. SPT Analyzer and core kit components.	
Trainee can explain pigtail and hardware-to-pigtail cable function.	
Trainee can identify connector care expectations and dielectric grease/dirt handling.	
Trainee can explain accelerometer vs strain gage function at a training level.	
Trainee knows strain gauges are not field-replaceable.	
Trainee can explain rod selection by confirmed AWJ/NWJ requirement.	
Trainee can list required software setup fields before data collection.	
Trainee understands automatic data folder storage at a training level.	
Trainee knows the PPE baseline and site-specific override rule.	
Trainee knows damage must be communicated to the office immediately.	

14. Future Confirmation Items

The module is ready for internal training use, but the following items should be confirmed later if the training program becomes a formal service manual or software workflow:

- Exact name/path of the designated automatic data folder on the analyzer hardware.
- Any customer-specific PPE exceptions beyond the Mobile Drill baseline.
- Formal company threshold for minimum observed calibrations and supervised data collections before solo work.
- Final version-controlled report handoff folder and naming standard once report workflow is formalized.

Revision History

Version	Status	Change
v0.1	Draft generated	Initial equipment/rig/hammer profile module created from extraction worksheet.
v0.2	Source-backed + SME reviewed	Updated with Pile Dynamics Inc. SPT Analyzer kit details, pigtail/cable path, instrumented rod sensor rules, file setup workflow, PPE baseline, automatic data folder language, accelerometer hardware, and equipment-damage escalation rule.